

Channel Geometry Analysis Technique for the Lower Mississippi River

Philip J. Soar¹, Colin R. Thorne², Oliver P. Harmar³, David S. Biedenbarn⁴ and C. Fred Pinkard⁴

¹*JBA Consulting, Magna House, South Street, Atherstone CV9 1DF, UK and Industrial Research Fellow, School of Geography, University of Nottingham, Nottingham NG7 2RD, UK*

²*School of Geography, University of Nottingham, Nottingham NG7 2RD, UK*

³*Halcrow Group Ltd., Arndale House, Otley Road, Leeds, LS6 2UL and Industrial Research Fellow, School of Geography, University of Nottingham, Nottingham NG7 2RD, UK*

⁴*Coastal and Hydraulics Laboratory, US Army Engineer Research and Development Center, ERDC, Vicksburg, MS 39180, USA*

26.1 INTRODUCTION

The channel of the Lower Mississippi River is primarily the product of natural flows acting on the floodplain materials over centuries and millennia to form an alluvial stream. However, the modern channel has been modified by engineered channel improvements performed as part of the Mississippi River and Tributaries (MR & T) project. Channel improvement features include meander cutoffs, levees, diversion structures, bank stabilization and low training dykes, which have been constructed to enhance flood control, stabilize the channel and provide sufficient depth for navigation even at times of low water (Moore, 1972). The form of the modern channel has been further influenced by morphological responses to artificial stabilization and training. Historically, morphological adjustments included bank erosion, channel siltation and recovery of some of the sinuosity lost due to artificial cutoffs, though more recently bank erosion, channel migration and planform evolution have all but ceased as revetments now extend along a great proportion of the banklines.

Substantive legislation for the MR & T project, and the initiation of the construction programme, began following

the catastrophic flood of 1927, and this event marked the beginning of a new era in terms of channel form and process in the Lower Mississippi River. Since that time, channel dimensions and geometries have evolved under the influence of natural flows, direct engineering interventions and dynamic adjustment driven by process-response feedback loops linking flow hydraulics and channel morphology.

Previous studies by the authors have documented the stage and slope adjustments on the Lower Mississippi River (Biedenbarn and Watson, 1997; Biedenbarn *et al.*, 2000), but the associated changes in channel geometry are only now being fully investigated (Harmar, 2004).

Channel geometry analysis is possible for the Lower Mississippi River thanks to the existence of historical, comprehensive and annual hydrographic surveys of the river. The comprehensive hydrographic surveys, made at approximately decadal intervals since the late nineteenth century, provide valuable information with respect to long-term channel changes during the twentieth century (Harmar *et al.*, 2005). However the comprehensive surveys only represent snapshots of channel form at the time when the survey was performed and local adjustments in channel geometry have been observed over much shorter, seasonal

timescales (Harmar, 2004). While both types of change are important, it is only the longer-term changes that represent the evolution of the river under the action of formative flows and in response to past and present engineering interventions. This evolution is now believed to be more complex than was indicated by analysis of water surface profiles alone (Biedenharn *et al.*, 2000; Harmar *et al.*, 2005). For this reason it is necessary to investigate changes in channel form based on analysis of the annual hydrographic surveys to elucidate the degree of spatial variability and the nature of short-term changes present in the river.

In this context, the research reported in this chapter concerns a pilot study performed in 2003 to ascertain the feasibility of using annual hydrographic survey data to investigate the channel geometry of the Lower Mississippi River. A test reach was selected between river miles 565 and 618 as the basis for detailed analysis of channel geometry and geometric changes that occurred during the 1990s. This reach was chosen as hydrographic survey data were readily available in electronic format. The aims were to develop a better appreciation of the inherent variability of channel geometry through time and space, and to provide improved geomorphological insights into dynamic adjustments that take place at the reach scale, over annual to decadal timescales. Only by combining knowledge of short-term variability with an understanding of longer-term process-response in the fluvial system will it be possible to separate local adjustments in channel geometry from wider-scale and longer-term channel evolution. Improved understanding of river forms and behaviour will provide a sound basis for future adaptive management and sustainable engineering of the river. As the research reported here was originally submitted to the US Army Engineer Research and Development Center, Vicksburg, the channel geometry analysis was undertaken using imperial units of measurements, although conversion to SI units has been made in the discussion below.

26.2 CONTEXT

26.2.1 Lower Mississippi River Channel Geometry

The Lower Mississippi River flows from the confluence of the Ohio River at Cairo, Illinois to the Gulf of Mexico, east of New Orleans, Louisiana (Figure 26.1). The river exhibits a meandering planform, first flowing through its alluvial valley and, farther downstream, across its deltaic plain. Prior to extensive engineering modification as part of the MR & T project, the Lower Mississippi River featured high rates of lateral channel mobility and was free to adjust its energy gradient through a combination of

gradual evolution and avulsive changes to its cross-sectional form and planform geometry. Adjustments were achieved through changes to the width, depth and bar morphology at pools and crossings that were integral to incremental growth and periodic cutoff of meander bends (Schumm *et al.*, 1994; Hudson and Kesel, 2000). Winkley (1977) reports that in its natural condition meander bend cutoffs developed quickly though not instantaneously along the river, and that, following a cutoff, the river took 30 to 80 years to regain its pre-cutoff width and bar sequencing. Moreover, cutoffs rarely occurred in multiples (with the exceptions of the occurrence of several cutoffs simultaneously following the New Madrid earthquake of 1811–12 and in response to riparian vegetation clearance along the Lower Mississippi River valley), but rather a long-term balance existed between reaches that were shortened by cutoffs and those being lengthened through meander growth.

During the twentieth century the geomorphology of the Lower Mississippi River was influenced by a series of engineering modifications to improve flood control and aid navigation. Channel slope was artificially steepened and sinuosity reduced by the construction of 14 artificial cutoffs in the reach between Memphis and the Old River tributary during the period 1932–1942 (Figure 26.1). Following the cutoff period, bank stabilization was undertaken along most of the length of the river using articulated concrete mattress and upper bank riprap (Moore, 1972). The effect has been to almost fix the relatively steep, low sinuosity, post-modification morphology of the river into a non-varying planform alignment. This is significant geomorphologically because the Lower Mississippi River can no longer adjust its energy slope through planform adjustments, but is, instead, restricted to making adjustments to its cross-sectional shape and longitudinal profile. MR & T work, as the name suggests, included measures on the tributaries as well as the Lower Mississippi itself. Improved land management practices in the catchment, together with construction of large reservoirs and channel improvement works on the Missouri and other large tributaries are believed by some researchers to have led to a dramatic reduction in suspended sediment load supplied to the Lower Mississippi River (Robbins, 1977). Reduction of upstream supply, coupled with the virtual elimination of sediment input from bank erosion, has been reported to have resulted in a marked reduction in the amount of sediment transported by the Lower Mississippi River (Keown *et al.*, 1981; Kesel, 1989; Chapter 9).

Recognition is growing that the geomorphological drivers and behaviour of the Lower Mississippi River have changed as a result of these engineering modifications (Smith and Winkley, 1996; Harmar and Clifford,



Figure 26.1 Location map showing the Lower Mississippi River, pilot study reach and gauging stations

2006). Accepting that geomorphological change (partly natural and partly in response to engineering interventions and catchment management) is occurring, inevitably leads to the conclusion that improved shorter-term and longer-term channel management strategies require an accurate understanding of how the river responded during the twentieth century and how it is likely to continue evolving during the twenty-first century.

Previously, morphological adjustments in the channel have been inferred from temporal variations in river stages

for selected discharges recorded at gauging stations along the Lower Mississippi River (Winkley, 1977). The discharge chosen for basis of comparison was usually that associated with the bankfull stage because this stage has been shown to approximate the dominant or channel-forming flow (Wolman and Miller, 1960; Hey 1975). Winkley (1977) concluded from temporal patterns in bankfull stages at gauging stations that, in the pre-cutoff period, the Lower Mississippi River was a stable system between Cairo, IL and Natchez, MS. Biedenharn and

Watson (1997) subsequently analysed specific gauge records for stations between Cairo and the Old River distributary in the post-cutoff period between 1950 and 1994 to propose a pattern of response through adjustments to the long profile at the regional scale (Figure 26.2).

Upstream of Arkansas City, the specific gauge records analysed by Biedenharn and Watson (1997) suggest a degradational trend during the late twentieth century, while downstream of Vicksburg the records suggest an aggradational trend. The absence of a significant trend in the reach between Arkansas City and Vicksburg has been interpreted as indicating that a 'hinge zone' of relative stability existed there during the late twentieth century. Hence, this model presents evidence to suggest that at the reach scale, the river responded to the cutoffs in a manner broadly similar to the type of morphological response associated with a single cutoff (as outlined by Lane, 1947).

Biedenharn and Watson's model provides a useful overview of geomorphological response, but its utility for future planning of river management is limited for two reasons. First, while specific gauge records reveal spatial and temporal variations in the water surface, these do not necessarily represent changes at the channel bed because changes in flow resistance are unaccounted for. Second, the simplicity of the model inevitably masks important details of reach and sub-reach scale morphological behaviour. These issues are explored further using the decadal

comprehensive hydrographic surveys by Harmar (2004) and Harmar *et al.* (2005). Schumm *et al.* (1994) have described the importance of neotectonic activity and geological factors in influencing pre-modification morphological behaviour at the reach and sub-reach scales. However, no study has investigated the continued importance of such influences on morphological behaviour during the post-modification period, and this is an urgent research need.

Hydrographic survey data collected during the late nineteenth and twentieth centuries can be used to improve understanding of long-term river behaviour because these data characterize morphological changes *directly* through examination of the bed topography, rather than indirectly through specific gauge records. Also, analysis of the survey data provides much finer spatial and temporal resolution than specific gauge records because surveyed sections are spaced a few tens or hundreds of metres apart while gauges are spaced at intervals of tens of kilometres along the channel.

26.3 DATA ACQUISITION AND PRE-PROCESSING

26.3.1 Pilot Study Reach

To explore the potential for channel geometry analysis using hydrographic survey data, a pilot study reach was selected between river miles 565 and 618 (Figure 26.1), for which data were readily available (river miles are counted upstream from the mouth of the Mississippi). The nearest gauging stations are Arkansas City at river mile 554.1, Rosedale at river mile 592.2 and Helena at river mile 663.1. The reach includes the confluences of the Arkansas and White Rivers with the Mississippi. Annual hydrographic surveys performed by the Vicksburg District, Corps of Engineers between 1992 and 2001 were obtained to represent channel variability and adjustment at a sub-decadal timescale. Cross-sections were surveyed at intervals of approximately 1000–1100 ft along the channel, with annual surveys repeated at the same fractional river miles. In general, the position of each cross-section varies by less than 100–150 ft, or 10% of the distance between successive cross-sections, which is considered to be an acceptable level of variability for this study.

26.3.2 Low Water Reference Plane

In extracting data to define the channel geometry from hydrographic surveys conducted on different dates, it is necessary to reference all elevations to a common datum.

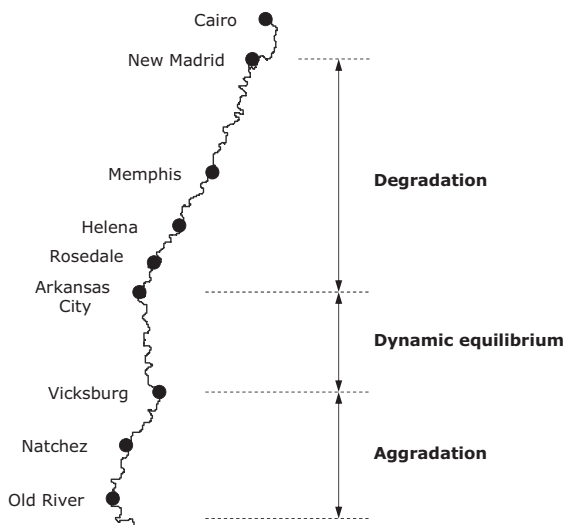


Figure 26.2 Regional scale adjustments to the longitudinal profile in the period 1950–1994, proposed by Biedenharn and Watson (1997) from analysis of specific gauge records

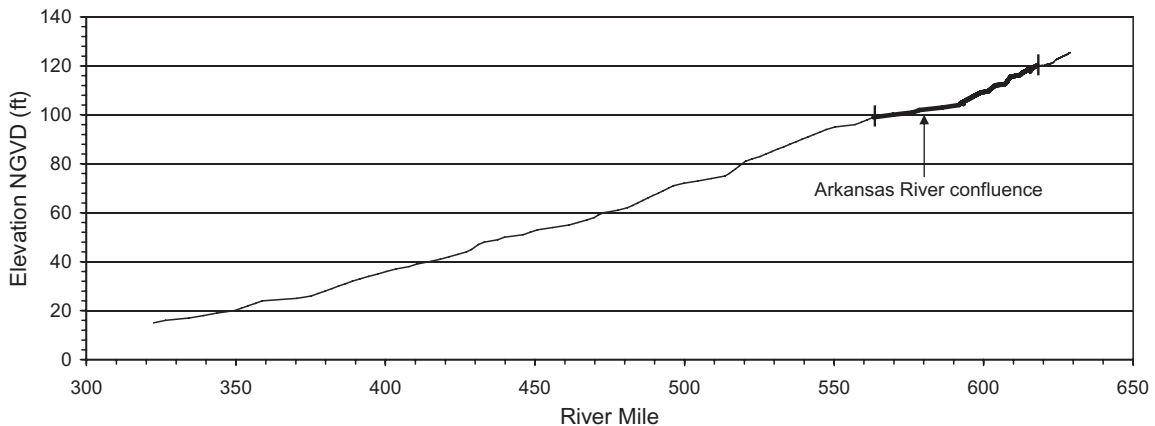


Figure 26.3 LWRP elevations in the Lower Mississippi computed by the Vicksburg District (including elevations computed by the Memphis District above RM 592). Thick line delimits the extent of the profile used in this study

In this study, the Low Water Reference Plane (LWRP) of the Mississippi River was used for this purpose. The LWRP was established by the US Army Corps of Engineers to set the crown elevations for dykes and other engineering works. It is also used by navigation interests as a guide to the depth of water available at critical locations in the river.

LWRP elevations along the Mississippi River are developed from stages computed at individual gauging stations based on the 97% exceedance flow for a specified period of record (typically from 1954 to the time of computation) being applied to a series of rating curves for a recent period (typically the past 10 years). This study used the most recent LWRP, which was computed in 1993 using the 1954 to 1991 flow period of record and 1982 to 1991 rating curves (Figure 26.3).

As this study has centred on an analysis of channel geometry above the LWRP, variability in the morphology of the bed due to bedforms, such as migrating dunes, cannot be discerned at this scale of analysis and is therefore not an objective of this study.

26.3.3 Separation of Bends and Crossings

A salient feature of most rivers, including the Lower Mississippi River, is local variability in the cross-sectional geometry of meander bends and crossings. It follows that useful additional information on the variability of channel geometry is obtained if cross-sections within bend reaches and crossing reaches are considered separately. Consequently, a channel planform curvature parameter was developed to discriminate between bends and crossings, with separate channel geometry analyses being performed

for each, in addition to the full data set of all cross-sections in each annual survey.

The technique assumes that meander bend and crossing 'reaches' can be suitably distinguished based solely on changes in channel curvature between successive cross-sections (Figure 26.4). The process adopted involved:

- (1) plotting the point in each cross-section with the lowest elevation, the thalweg (not the centreline, as there are marked discontinuities along the study reach where the length of cross-section changes markedly over a very short distance);
- (2) smoothing the line using MapInfo GIS software (particularly within the crossing reaches where the deepest point in the channel is not always clearly defined);
- (3) applying a macro to calculate the absolute (positive) change in channel direction between cross-sections;
- (4) applying a smoothing algorithm using Tecplot data processing and visualization software to define similar reaches.

The approach revealed that bend and crossing sub-reaches tend to be distinguished by a threshold of about 5° of change in channel direction between consecutive cross-sections. The final classification scheme therefore defined bend reaches as having a change in direction greater than 6° , crossing reaches less than 4° change, and transitional reaches between 4° and 6° change (Figure 26.5).

26.3.4 Divided Channels

While the modern Lower Mississippi River predominantly has a single-thread, meandering channel, it does exhibit

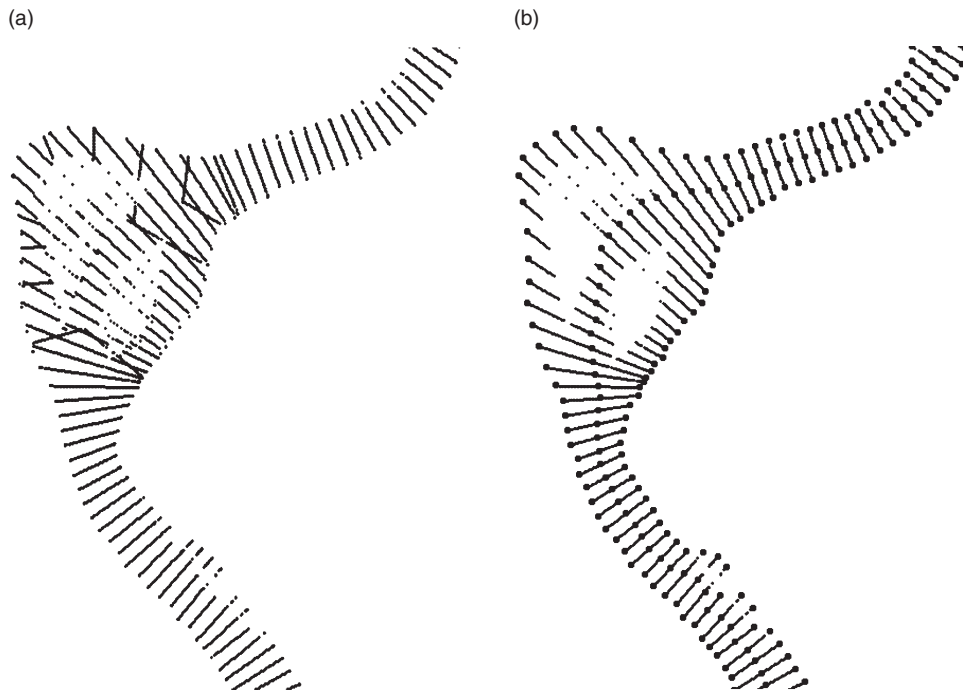


Figure 26.4 Example of initial data processing: (a) original cross-sections plotted in ESRI Arc View 3.3 GIS and (b) processed cross-sections following initial removal of 'problem' data and application of automated routines (Harmar, 2004)

some reaches with divided channels. These are often found in asymmetrical cross-sections with wide point bars and chute channels. Clearly, the cross-sectional geometry of divided reaches with secondary channels differs to that of single-thread reaches. Given the ecological importance of secondary channels, information on the geometry of such reaches is of interest. Care was therefore taken to identify cross-sections located within reaches with divided channels and treat them separately to explore the possibility of defining separate channel geometry analyses for these reaches.

To meet this objective, where secondary channels were present in a cross-section, the width, average depth, etc., were calculated for both the entire cross-section (all channels added together) at the specified elevation and then for just the primary channel, defined as that part of the cross-section containing the lowest elevation. The effect on the channel geometry analysis was, however, found to be minor when considering the reach as a whole and only very localized when comparing the geometry of individual cross-sections, because secondary channels are uncommon and, where they do exist, they tend to become included within the primary channel at elevations at or about LWRP.

26.3.5 Pre-Processing Procedure for Hydrographic Survey Files

Pre-processing of the hydrographic survey data was undertaken at the University of Nottingham as part of a doctoral research project (Harmar, 2004). The objective of pre-processing was to develop a series of automated routines for extracting survey data pertaining to individual cross-sections as stand alone text files from the full data set of each annual survey, originally compiled in GIS format.

Each annual hydrographic survey file for the period 1992–2001 was obtained in Microstation design file format (.dgn) from the USACE Vicksburg District Office, Vicksburg, MS. Four stages of data pre-processing were then applied:

- (1) extraction of xyz hydrographic coordinates into a standard text file;
- (2) identification of individual cross-sections;
- (3) sorting of cross-sections into the downstream order;
- (4) rectifying the cross-section points into straight lines.

Once pre-processing was complete, a final visual inspection was undertaken. This involved displaying the com-

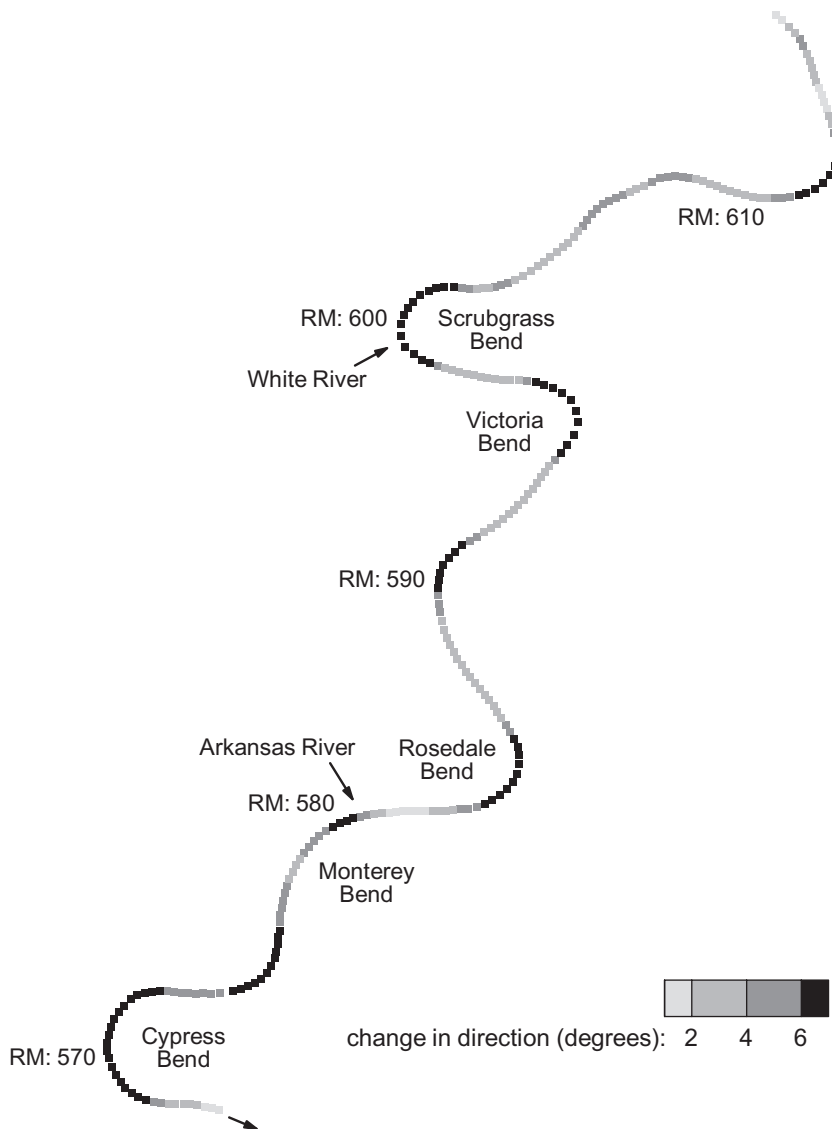


Figure 26.5 Change in channel direction (positive) between cross-sections

puted centre-point and associated left and right bank coordinates for each cross-section in a GIS layer (Figure 26.4).

26.3.6 Data Projection

Corpscon (Corps conversion) software, developed by the US Army Topographic Engineering Center (TEC), was used to convert all cross-section point coordinates to the regional standard, NAD 1983, Universal Transverse Mer-

cator (UTM) Zone 15 system. All elevations are given in US customary units and are referenced to the National Geodetic Vertical Datum of 1929 (NGVD29).

26.3.7 Cross-Section Screening

The first task in processing the cross-sectional data was to 'screen' the cross-sections and determine how far above LWRP each one extends. This was essential to check the feasibility of analyzing channel geometry distributions for

Table 26.1 Results of initial screening of cross-sections

Year	<LWRP	>LWRP	>LWRP + 5 ft	>LWRP + 10 ft	>LWRP + 15 ft	>LWRP + 20 ft	>LWRP + 25 ft	>LWRP + 30 ft	Total
1992	188	68	51	34	19	10	0	0	256
1993	196	60	25	8	0	0	0	0	256
1994	126	153	119	62	6	1	0	0	279
1995	196	60	25	6	0	0	0	0	256
1996	36	220	185	5	5	0	0	0	256
1997	23	233	230	226	213	194	81	6	256
1999	127	132	79	48	5	0	0	0	259
2000	245	11	0	0	0	0	0	0	256
2001	50	205	184	163	131	100	87	64	255

Table 26.2 Revised screening of cross-sections, following limited extrapolation

Year	<LWRP	>LWRP	>LWRP + 5 ft	>LWRP + 10 ft	>LWRP + 15 ft	>LWRP + 20 ft	>LWRP + 25 ft	>LWRP + 30 ft	Total
1992	159	97	67	50	31	17	7	0	256
1993	158	98	63	27	6	0	0	0	256
1994	107	172	153	119	66	14	1	1	279
1995	141	115	77	35	5	0	0	0	256
1996	30	226	201	175	36	5	4	0	256
1997	14	242	233	230	224	203	190	115	256
1999	60	199	158	82	48	11	1	0	259
2000	126	130	35	0	0	0	0	0	256
2001	41	214	208	187	160	134	105	91	255

the separate years and the elevations proposed. The results (Table 26.1) indicate that the vertical extent of cross-sections varies widely between years. For example, the dataset for the year 2000 is of limited use as this was a very low flow year and the survey was only undertaken for the wetted perimeter. In contrast, cross-sections for high water years, such as 1992, 1994, 1997 and 2001, extend to higher elevations and cover a greater range of elevations above LWRP.

Initial screening adopted a strict approach whereby a cross-section was excluded if either of its end points fell below the specified elevation, even by a small amount. In this initial process, many cross-sections that fell just short were excluded by this criterion. A sensitivity analysis was performed to investigate whether limited linear extrapolation of cross-sections would improve matters significantly. The final scheme adopted was to allow extrapolation of the bank for a vertical distance of 3.05 m (10 ft) if the bank slope (vertical to horizontal) was greater than 0.2 (which corresponds to banks covered by revetments, where it may be assumed that the steep slope of the bank is likely to continue to a higher elevation) and extrapolation through a vertical distance of 1.52 m (5 ft) if the bank slope was between 0.025 and 0.2. Cross-sections with flatter side

slopes were not extrapolated at all, due to uncertainty concerning the topography of low angle, shelving banks. The revised cross-section screening results are listed in Table 26.2.

The outcome of the extrapolation process varies between years, but overall it assisted in developing a substantially larger dataset for undertaking channel geometry analysis. At the LWRP elevation, extrapolation has had a negligible impact in 1996, 1997 and 2001 (the latter two years being high water years) but a large impact in 1995, 1999 and 2000. However, at higher elevations, the impact in the higher water years was marked, with, for example, an additional 170 cross-sections included in 1996 for 3.05 m (10 ft) above LWRP and an additional 109 cross-sections included in 1997 for both 7.62 m (25 ft) and 9.14 m (30 ft) above LWRP.

26.4 ANALYTICAL APPROACH AND METHODOLOGY

26.4.1 Channel Geometry Analysis

For each cross-section, the width, average depth, maximum depth, wetted perimeter, hydraulic radius and cross-section

area were calculated at the water surface elevation corresponding to LWRP and at 0.30 m (1 ft) increments above this, up to LWRP + 9.14 m (30 ft), which corresponds approximately to bankfull stage identified by Biedenharn and Thorne (1994), or once the elevation was greater than either of the cross-section end points. This involved two analytical stages:

First, a macro was used to prepare the data into a suitable format for calculating the above parameters. This macro included:

- (1) extending the cross-section according to the criteria previously discussed;
- (2) calculation of the distance between cross-section points;
- (3) calculation of the elevation difference between the interpolated LWRP and both cross-section end points and then recording the lowest. This became the largest 'Elevation above LWRP' for which the channel geometry analysis was undertaken (and also facilitated the development of Tables 26.1 and 26.2);
- (4) calculation of the Easting and Northing coordinates of the deepest point in each cross-section (used to calculate changes in channel direction for defining bend and crossing reaches).

The second stage involved calculation of the channel geometry parameters. A further macro was used to undertake the calculations and involved dividing each cross-section into vertical panels, calculating the top width, wetted perimeter (for channel boundaries only) and cross-section area of each panel and then summing these values to find the cross-section totals. The average depth was defined as the ratio of the cross-sectional area to the total width and the hydraulic radius as the ratio of the cross-section area to the wetted perimeter. The procedure was applied to the main channel only in cross-sections with a secondary channel (see discussion on divided reaches above).

26.4.2 Probability Analysis

Conventional hydraulic geometry analysis involves developing power-law relations to predict various channel geometry parameters. However, examination of the variation of width, average depth, etc., with elevation above LWRP did not reveal the type of close association that is normally required to define a best-fit regression line. Instead, for a given elevation above LWRP, it was apparent that there are wide-ranging values of width, depth, etc., and fitting any type of best-fit line would not only be questionable statistically, but would also obscure the

inherent variability in channel geometry that is an important characteristic of the Lower Mississippi River. Clearly, an alternative technique was required to 'capture' this variability, display it and represent how it changes with elevation above LWRP, downstream location and between annual surveys, without any loss in information.

For each annual dataset, the approach adopted was to plot elevation above LWRP against channel geometry parameter for each cross-section on the same graph. Then, each data point on the graph was assigned a cumulative probability (from 0 to 1), which can be interpreted as the probability of the channel geometry parameter being less than or equal to that particular data value, at the chosen elevation above LWRP (Stage 1). These probabilities were then rectified to a regular grid by linear interpolation, first along the y-axis for each 0.30 m (1 ft) increment above LWRP (Stage 2) and then across the remainder of the grid (Stage 3). This procedure is illustrated for width in Figure 26.6, which also demonstrates the high degree of variability that has warranted this probabilistic approach. With each annual dataset, the probability analysis was repeated for each channel geometry parameter, firstly for all cross-sections in the study reach, secondly for those cross-sections located in pool sub-reaches and thirdly for those cross-sections located in crossing sub-reaches.

To avoid misleading representations of probability distribution associated with very small sample sizes, cumulative probabilities for each elevation above LWRP were calculated only if they contained at least 10 cross-sections that extended above that elevation. Thus, many of the final plots do not extend as far as 4.57 m (15 ft) above LWRP and very few go all the way up to 9.14 m (30 ft) above LWRP. Finally, the 50th percentile line was calculated and plotted on the cumulative graphs to illustrate how the median channel width varied with stage for that survey date. A series of Visual Basic macros and Tecplot processing and visualization routines were used to undertake these processes.

26.4.3 Spatial Analysis

Spatial analysis was undertaken to explore how channel geometry varied along the study reach at the time of each hydrographic survey. The approach involved plotting the location of each cross-section as a series of symbols on a map of the river in the study reach and colouring the symbols according to the value of width, average depth, etc.

As the precise locations of the cross-sections established in each of the hydrographic surveys vary only slightly, for visualization purposes (and to facilitate the temporal analysis described below) the 2001 survey was

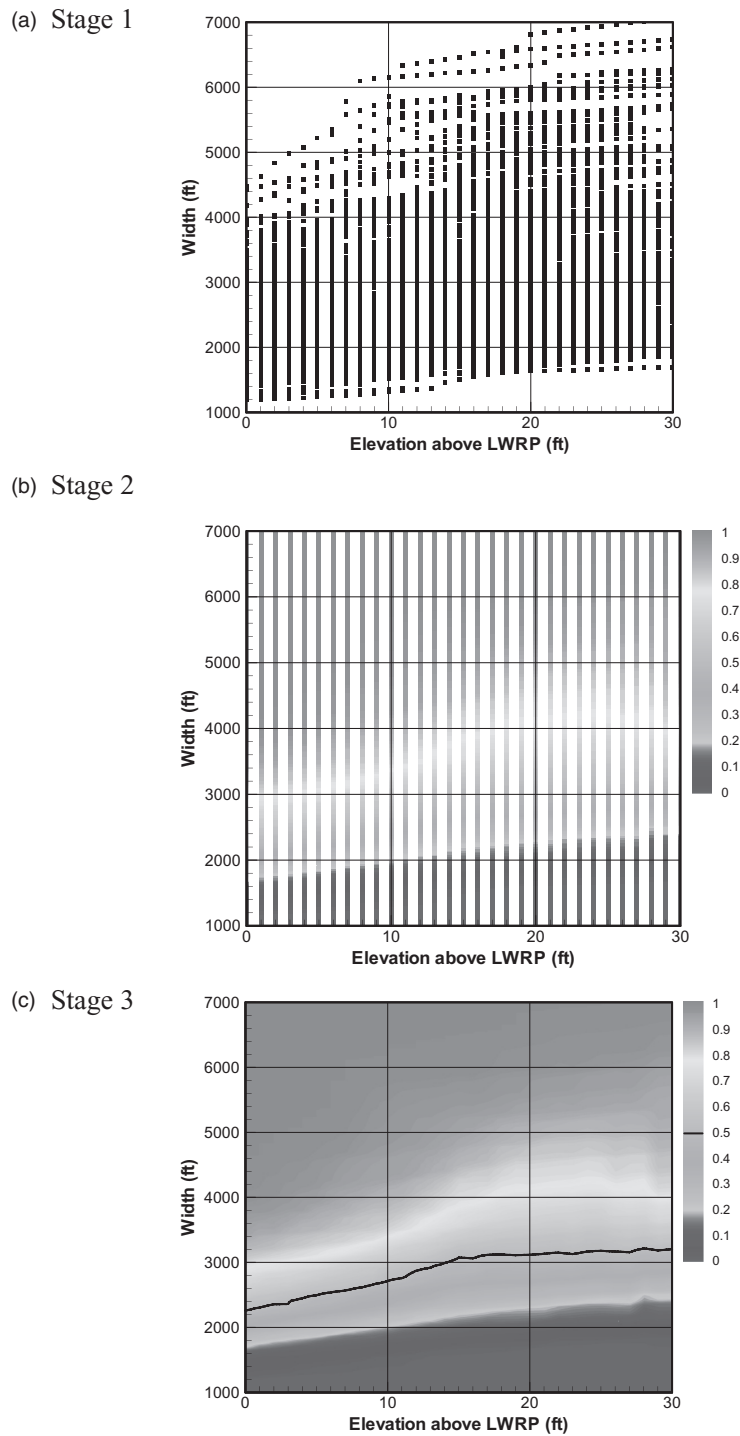


Figure 26.6 (See also colour plates.) Development of cumulative frequency plots of channel geometry variability with elevation above LWRP. (a) Stage 1: plotting data values for each cross-section. (b) Stage 2: assigning probability (0 to 1) and interpolation along the y-axis. (c) Stage 3: interpolation of probability across the remainder of the grid, showing the 50% probability line. The example shown is for width in the 1997 survey

used to represent the location of each cross-section for all years. Values for cross-sections that could not be calculated (because one or both end points did not extend above the specified elevation) were interpolated from neighbouring sections. If less than 10 cross-sections were available, then no values were shown in the final plots as results for the specified elevation above LWRP were unsuitable for spatial analysis due to the very small sample size.

26.4.4 Temporal Analysis

Temporal analysis was performed to reveal fluctuations and/or trends of change in channel geometry during the 1990s. The analysis used the same graphical format as the spatial analysis, but with the cross-section symbols coloured according to the direction and magnitude of change in average depth and change in cross-section area between annual surveys and over the period of observation, 1992–2001. As all parameters are referenced to a fixed datum (LWRP), changes in average depth are indicative of aggradation/degradation within the study reach, while changes in cross-section area highlight zones of scour and fill.

The temporal analysis was only undertaken for the LWRP elevation as the large number of interpolated values at higher elevations could generate misleading patterns of channel change.

26.5 RESULTS

26.5.1 At-a-station Channel Geometry

Archived results

The database and results generated by the probability, spatial and temporal analyses are not amenable to publication in printed format because there are, in total, 1305 plots (324 for the probability analysis, 945 for the spatial analysis, and 36 for the temporal analysis). Instead they can be obtained by contacting the first author at JBA Consulting or the second author at the University of Nottingham. The archive contains all the cross-sectional data, with Eastings and Northings referenced to the NAD 1983, Universal Transverse Mercator (UTM) Zone 15 coordinate system. Elevations are given in US customary units (feet and tenths) as these are the units in which the original observation were made, and are referenced to the National Geodetic Vertical Datum of 1929 (NGVD29). The main functionality of the data archive is its interactive browser facility which permits users to navigate through the plots and scroll through different years to examine

central tendencies, variability and changes in channel geometry at will. Complete instructions for using the browser are provided.

Results for at-a-station channel geometry

Sample results from the probability analysis used to display at-a-station channel geometry are illustrated in Figure 26.7. Each plot in Figure 26.7 represents data from a single annual hydrographic survey and for a single channel geometry parameter. The first two plots represent channel geometry for cross-sections located firstly in bend reaches and secondly in crossing reaches. The full set of results is available in the interactive archive.

26.5.2 Spatial Variability and Adjustments

Sample results from the spatial analysis to examine how channel geometry parameters vary along the study reach are illustrated in Figure 26.8. The full set of results is available in the interactive archive.

26.5.3 Temporal Variability and Adjustments

Sample results from the temporal analysis, used to examine how average depth and cross-section area within the study reach change between years and over the period of record (1992–2001), are illustrated in Figures 26.9 and 26.10. The results are relative to LWRP, which provides a time-invariant reference plane. Aggradation and degradation can be inferred from the changes in average depth shown in Figure 26.9, while scour and fill can be inferred from changes in the cross-sectional area shown in Figure 26.10. The full set of results is available in the interactive archive.

26.6 INTERPRETATION AND COMMENTARY

26.6.1 Channel Geometry Analysis

With over 1300 plots in the results database, the following commentary summarizes the channel geometry characteristics for the 1992–2001 period as a collective dataset only and differences between annual surveys are reserved for the discussion on temporal variability.

Width

The results indicate that, in the study reach, channel width at LWRP fluctuates around a value of 670 m (2200 ft) $\pm$ 61 m (200 ft), with bends ranging 625–823 m (2050–2700 ft) being generally wider and more variable in width

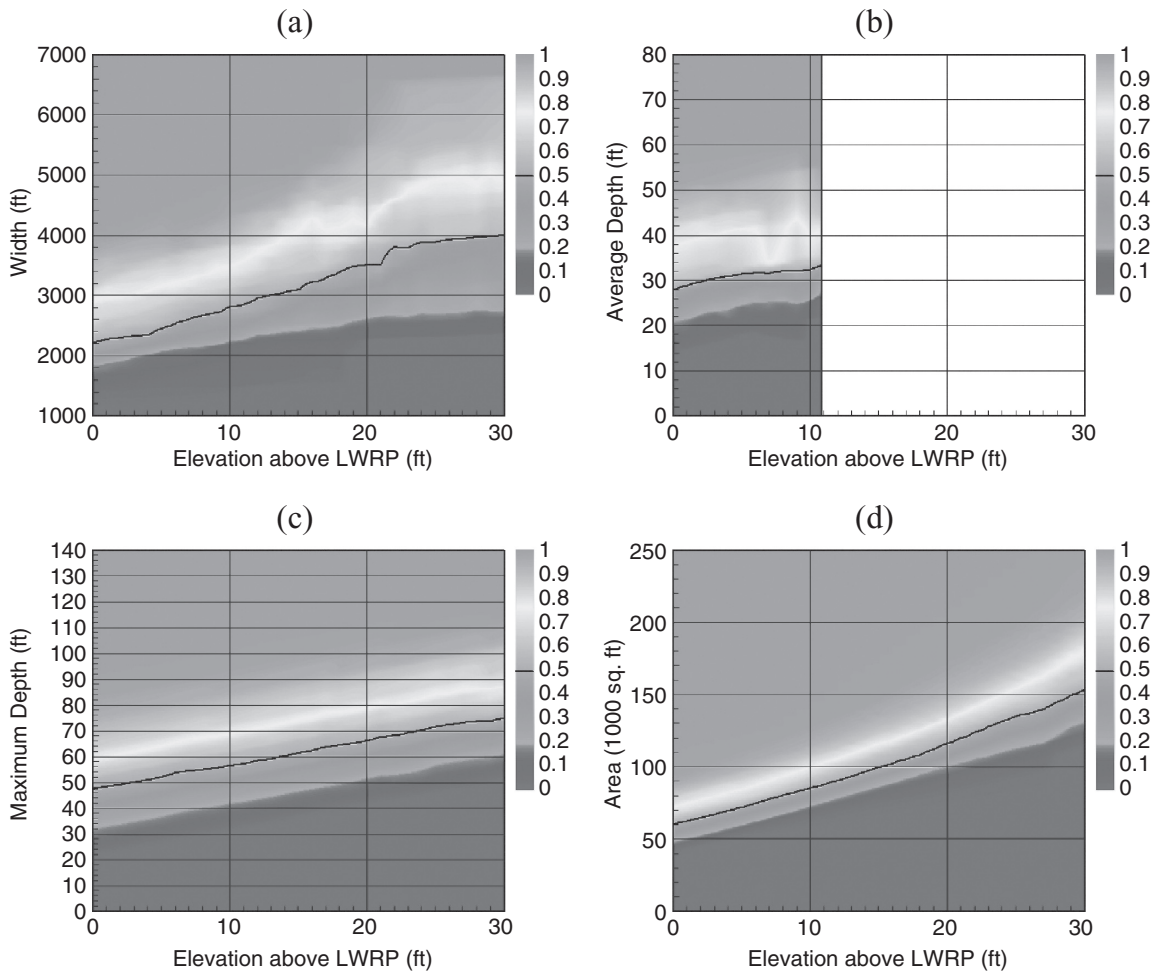


Figure 26.7 (See also colour plates.) Sample results from the probability analysis: (a) width in 2001 for bends; (b) average depth in 1995 for crossings; (c) maximum depth for all sections in 2001; (d) cross-sectional area in 1997 for all sections

than crossings ranging 533–670 m (1750–2200 ft). Channel width appears to increase with stage more rapidly at bends than at crossings. At high stages [based on LWRP + 6 m (20 ft)], channel widths in bends and crossings converge and are consistently around 945 m (3100 ft). Unfortunately, due to limitations in the hydrographic surveys, extrapolation to the bankfull stage is not possible.

Average depth

Channel average depth at LWRP fluctuates about 8.5 m (28 ft), ranging between 7.6 and 9.8 m (25 and 32 ft). Bend and crossing average depths are generally similar, although the inter-quartile range for crossing average depths is

usually wider than that for bends. In high water years, average depths are approximately equal at bends and crossings. The high flow channel average depth [based on LWRP + 6.10 m (20 ft)] is usually 10 to 11 m (33 to 37 ft).

Maximum depth

The relationship between maximum depth and elevation above LWRP should be linear by definition. Deviations from linearity present in the results reflect data limitations. They occur because as stage increases, the number of cross-sections with data decreases, producing changes in the sampled data and nonlinearity in the median line.

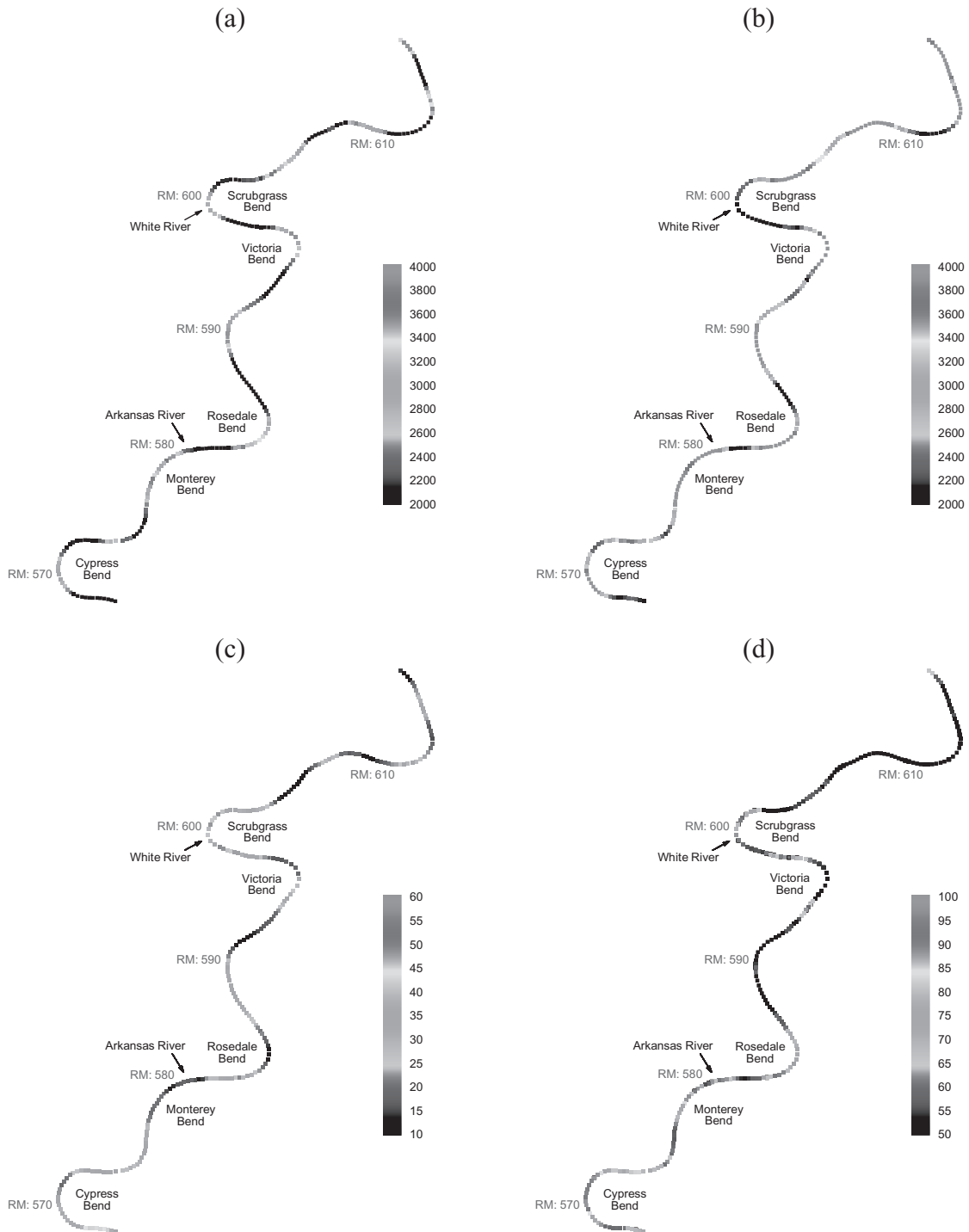


Figure 26.8 (See also colour plates.) Example results from the spatial analysis: (a) width (ft) in 1997 at LWRP; (b) width (ft) in 1997 at 20 ft above LWRP; (c) average depth (ft) in 2001 at LWRP; (d) area (1000 ft²) in 1996 at LWRP (this plot also shows interpolated values in white)

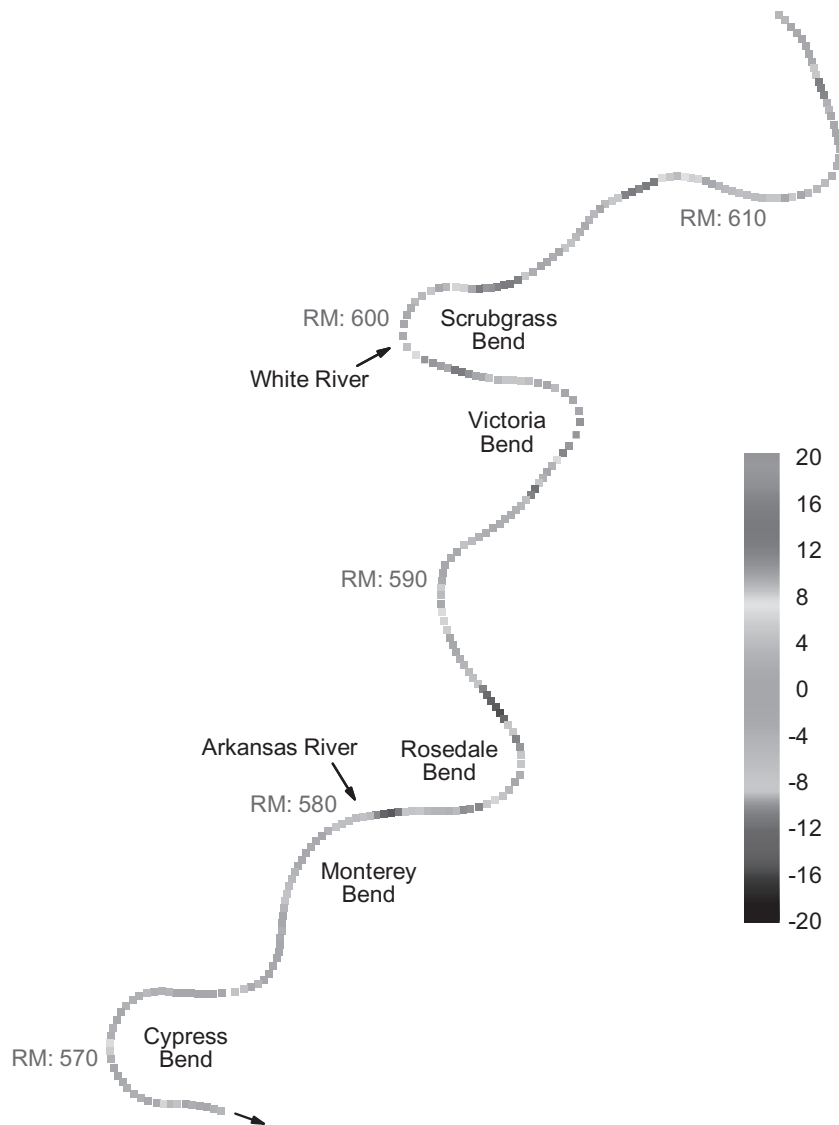


Figure 26.9 (See also colour plates.) Changes in average depth (ft) between 1992 and 2001, indicative of aggradation and degradation

Channel maximum depth at LWRP fluctuates about 15.9m (52 ft), ranging between 14 and 16.4m (46 and 54 ft). Bend and crossing maximum depths are generally similar, although bend maximum depths at LWRP are more variable. The lack of difference between bend and crossing maximum depths may be explained by the fact that what is plotted is the median maximum depth for several cross-sections and the fact that the deepest point in a bend pool may not actually lie within the planimetric

bend, being located in the next crossing reach downstream. The high flow channel-averaged maximum depth [based on LWRP + 6.10m (20 ft)] is around 20 to 21.9m (66 to 72 ft).

Wetted perimeter and hydraulic radius

Wetted perimeter and hydraulic radius are indistinguishable from width and average depth respectively, due to

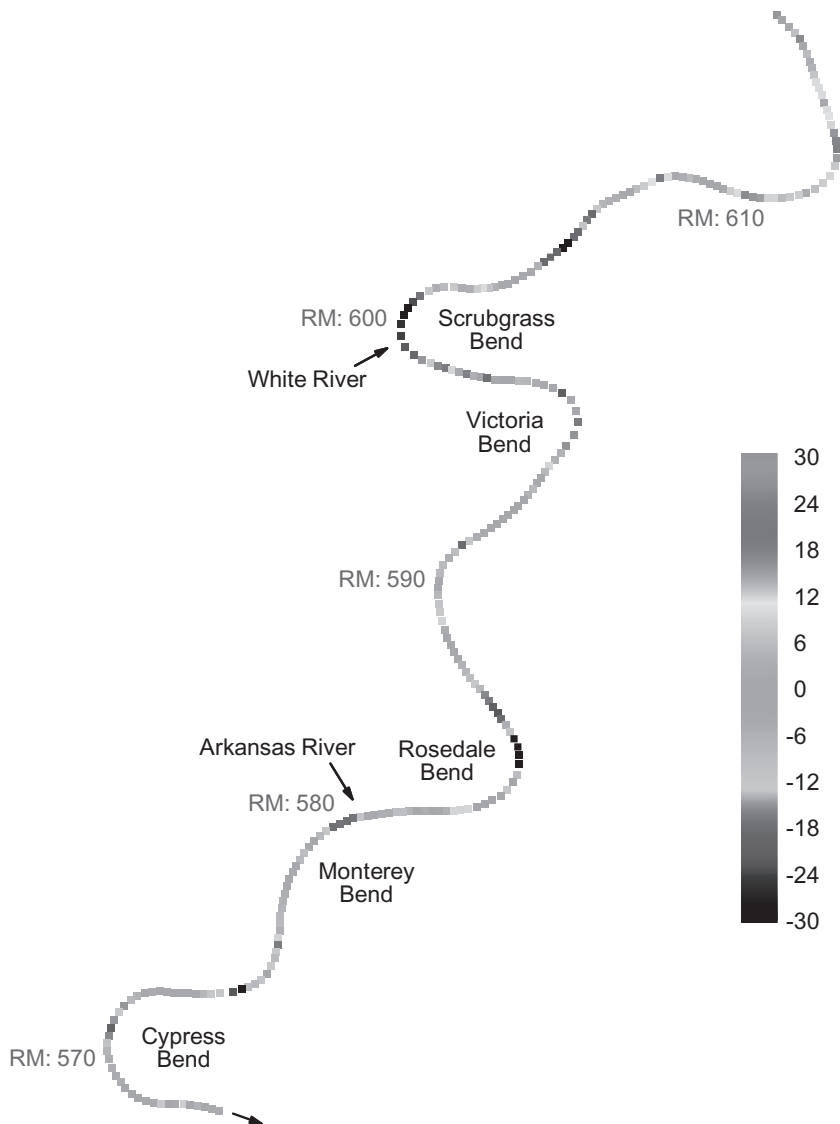


Figure 26.10 (See also colour plates.) Changes in area (1000 ft²) between 1992 and 2001, indicative of scour and fill

the high width–depth ratio of the channel of the Lower Mississippi River.

Cross-sectional area

Channel cross-sectional area at LWRP fluctuates about 5574 m² (60 000 ft²), ranging between 5110 and 6039 m² (55 000 and 65 000 ft²). Bends tend to have larger and more variable areas than crossings. This difference may

be explained by differences between cross-section shapes at bends and crossings. Compared with crossings, bend reaches tend to be less uniform, less trapezoidal and more variable in depth across their point bars.

At high flow [based on LWRP + 6.10 m (20 ft)], the channel area is around 10 219 to 11 613 m² (110 000 to 125 000 ft²) again with bends ranging 11 148 to 11 613 m² (120 000 to 125 000 ft²) having larger areas than crossings between 10 219 to 11 148 m² (110 000 to 120 000 ft²). The

inter-quartile ranges for cross-sectional area at both bends and crossings are narrow.

26.6.2 Spatial Analysis

Width varies locally between bends and crossings, but there are no marked reach-scale differences. Average depths display no reach-scale trend. The greatest average depths are found at bend exits or in the upstream parts of crossing reaches (just downstream of bend exits). The cross-sections with the greatest maximum depths are located at bend exits or upstream of the inflection points in crossing reaches and no reach-scale spatial variation can be discerned.

26.6.3 Temporal Analysis

There may be a tendency for the channel at LWRP to widen between 1992 and 1997, but to be narrower in 2001 (especially at Scrubgrass Bend). At the reach-scale, average depth changes little between 1992 and 2001. About 75% of changes are less than ± 1.5 m (5 ft) in magnitude. Changes that are apparent indicate a tendency for bed scour at and downstream of bend exits. For example, average depth increases in excess of 3 m (10 ft) at Rosedale, Monterey and Cypress bends in 1993–1994, and aggradation on the crossings downstream.

Examination of changes on a year to year basis reveals only short-term adjustments that are not sustained over the longer period. Considerable activity occurs in the period

1993–1994, possibly related to the higher volume of runoff experienced in 1994 (high water year), compared with the lower volume in 1993 (as illustrated in Figure 26.11, based on flow data from the Vicksburg gauging station). In addition, there was over 6 m (20 ft) of bed scour at Victoria Bend between 1997 and 1999, which may be a local adjustment or could be a response to bendway weirs that were installed at that bend in 1995, and which did not initially operate as intended.

The absence of reach-scale changes in average depth indicates a lack of aggradation or degradation that is consistent with the reach being dynamically stable during the 1990s, as indicated by specific gauge analyses performed by Biedenharn and Watson (1997).

Examination of cross-sectional areas reveals that between 1992 and 2001 there were notable reductions in channel area around the Scrubgrass and Rosedale Bends. At both bends, the channel has filled by more than 2322 m² (25 000 ft²). Crossings generally changed by less than ± 743 m² (8000 ft²) between 1992 and 2001.

In terms of year on year changes, there is little activity in 1992–1993, but in 1993–1994 there was more scour and fill. This did not, however, follow any systematic pattern related to bends and crossings. Scour and fill in 1994–1995 tended to reverse the changes that occurred during the previous period. There was little activity in 1995–1996, 1996–1997 and 1997–1999. Apparent activity at Victoria Bend in 1999–2000 and at other locations in 2000–2001 is probably due to lack of data in this area for 2000, which was a low runoff year.

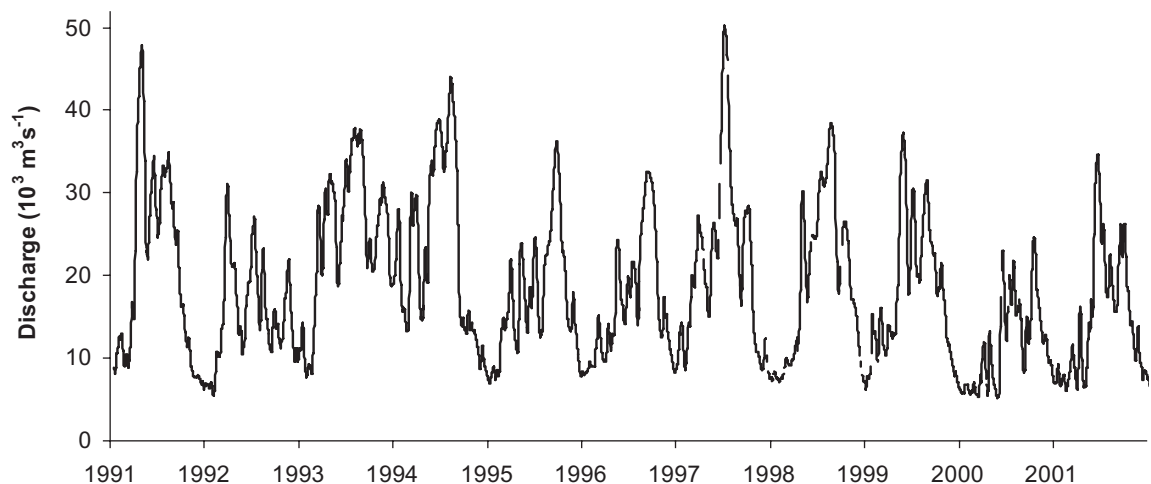


Figure 26.11 Discharge recorded at Vicksburg between 1991 and 2001

26.7 CONCLUSION

The channel geometry analysis performed in this project has showed the viability of extracting useful morphological information from historical, annual hydrographic surveys. The strength of the probabilistic technique developed to analyze and display the channel geometry data extracted from the hydrographic surveys lies in the fact that the results preserve all of the original data and demonstrate that the cross-sectional geometry of the Lower Mississippi River is highly variable in both time and space. Despite this, the fact that median values of key at-a-section parameters such as width, average depth, maximum depth, and cross-sectional area increase monotonically with increasing stage gives an insight into the cross-sectional morphology of this extensively engineered river.

Morphological diversity of the type evident in the Lower Mississippi River is a property inherent in natural channels that is often lost when river re-sectioning or training works are designed on the basis of regression equations describing the stable hydraulic geometry of regime channels. Application of engineering designs that use regime-style equations for dynamic stability as templates, while allowing for natural variability that mimics that displayed by the study reach, may provide more sustainable solutions to channel instability problems (Soar and Thorne, 2001).

Temporal changes in cross-sectional geometry demonstrate that the behaviour of this reach is highly dynamic. Examination of the full archived dataset shows that year-on-year changes observed in the study reach are larger than the net change observed over the decadal period of record, and there is some evidence that morphology responds to the magnitude of annual runoff. However, even a decade long period of record is short for a river as large as the Lower Mississippi and no firm conclusion concerning the adjustment status of this reach can be drawn from the analysis presented herein.

Establishing the historical and current status of morphological channel adjustment would be very useful in decision-making concerning engineering measures to deal with sediment and morphology-related problems in this sub-reach and this is the subject of parallel research by the authors that uses a multi-scaled approach to examine morphological changes at the sub-reach, reach, and system scales (Harmar *et al.*, 2005).

The pilot analysis performed in this study covered a sub-reach that is only a fragment of the Lower Mississippi River. It would be highly informative to apply the channel geometry analysis technique to other selected reaches, within the context of channel stability analyses at the local, reach, and system scales, in order to better under-

stand the response of the Lower Mississippi to past engineering interventions in the fluvial system.

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